

DEPARTMENT OF MATHEMATICS
BIRLA INSTITUTE OF TECHNOLOGY MESRA, RANCHI
MA24101 Mathematics-I, Session: (MO-2025)
Assignment - 5 (Module V)

1. If $\vec{A} = 5t^2\hat{i} + t^3\hat{j} - t\hat{k}$ and $\vec{B} = 2(\sin t)\hat{i} - (\cos t)\hat{j} + 5t\hat{k}$, find
 - a) $\frac{d}{dt}(\vec{A} \cdot \vec{B})$
 - b) $\frac{d}{dt}(\vec{A} \times \vec{B})$
2. Find a vector normal to the surface at the given point
 - a) $f(x, y) = y \ln x + xy^2$
 - b) $f(x, y, z) = 2z^3 - 3(x^2 + y^2)x + \tan^{-1}(xz)$ at $(1, 1, 1)$
 - c) $f(x, y, z) = e^{x+y} \cos z + (y + 1) \sin^{-1}(x)$ at $(0, 0, \frac{\pi}{6})$
3. Find the constants a and b so that the surface $ax^2 - byz = (a + 2)x$ is orthogonal to the surface $4x^2y + z^3 = 4$ at the point $(1, -1, 2)$.
4. Find the directional derivative of the function at the given point P_0 in the direction of the vector $\vec{A}$
 - a) $f(x, y) = 2xy - 3y^2$, $P_0(5, 5)$, $\vec{A} = 4\hat{i} + 3\hat{j}$
 - b) $f(x, y, z) = 3e^x \cos(yz)$, $P_0(0, 0, 0)$, $\vec{A} = 2\hat{i} + \hat{j} - 2\hat{k}$
5. Find the direction in which the functions increase and decrease most rapidly at the given point P_0 . Find also the directional derivative of the function in that direction
 - a) $f(x, y) = x^2 + xy + y^2$, $P_0(-1, 1)$
 - b) $f(x, y, z) = \ln(x^2 + y^2 + 1) + y + 6z$, $P_0(1, 1, 0)$
6. Find the directional derivative of $f(x, y, z) = x^2yz^3$ along the curve $x = e^{-u}$, $y = 2 \sin u + 1$, $z = u - \cos u$ at the point P where $u = 0$
7. In what direction from $(3, 1, -2)$ is the directional derivative of $\phi = x^2y - 2x^4$ maximum. Find also the magnitude of this maximum.
8. Evaluate $\text{div } R$ and $\text{curl } R$ and $\text{div}(\text{curl } R)$ where
 - a) $\vec{R} = (x^2y^3 - z^4)\hat{i} + 4x^5y^2z\hat{j} - y^3z^6\hat{k}$
 - b) $\vec{R} = (x - y)^3\hat{i} + e^{yz}\hat{j} + xye^{2y}\hat{k}$
9. Find the work done in moving a particle once around a circle C in the XY plane, of the circle has centre at the origin and radius 2 and if the force field is given by $\vec{F} = (2x - y + 2z)\hat{i} + (x + y + z)\hat{j} + (3x - 2y - 5z)\hat{k}$.
10. Find the circulation of F around the curve C where $\vec{F} = y\hat{i} + z\hat{j} + x\hat{k}$ and C is the circle $x^2 + y^2 = 1$, $z = 0$.
11. Find the work done by the force If $\vec{F} = yz\hat{i} + xz\hat{j} + xy\hat{k}$ acting along the curve given by $\vec{R} = t^3\hat{i} + t^2\hat{j} + t\hat{k}$ from $t = 1$ to $t = 3$.
12. Show that $\vec{F} = (2xy + z^3)\hat{i} + x^2\hat{j} + 3xz^2\hat{k}$ is a conservative force field. Find the scalar potential. Find also the work done in moving an object in this field from $(1, -2, 1)$ to $(3, 1, 4)$.

13. Evaluate $\int_{(0,0)}^{(2,1)} (10x^4 - 2xy^3)dx - (3x^2y^2)dy$ along the path $x^4 - 6xy^3 = 4y^2$.
14. Applying Green's theorem to evaluate $\oint_C e^{2x} \sin(2y)dx + e^{2x} \cos(2y)dy$, where C is the ellipse $9(x - 1)^2 + 4(y - 3)^2 = 36$.
15. Applying Green's theorem to evaluate $\oint_C (x^5 + 3y)dx + (2x - e^{y^3})dy$, where C is the circle $(x - 1)^2 + (y - 5)^2 = 4$.
16. Verify Green's theorem $\oint_C (xy + y^2)dx + x^2dy$, where C is bounded by the curve $y = x, y = x^2$.
17. Find the surface area of the portion of the cylinder $x^2 + z^2 = 4$ lying inside the cylinder $x^2 + y^2 = 4$.
18. Find the surface area of the portion of the sphere $x^2 + y^2 + z^2 = 9$ lying inside the cylinder $x^2 + y^2 - 3y = 0$.
19. Evaluate $\iint_S yz\hat{i} + zx\hat{j} + xy\hat{k}ds$, where S is the surface of the sphere $x^2 + y^2 + z^2 = a^2$ in the first octant.
20. By Gauss's Divergence theorem evaluate $\iiint_S x^2dydz + y^2dzdx + 2z(xy - x - y)dxdy$ where S is the surface of the cube $0 \leq x \leq 1, 0 \leq y \leq 1, 0 \leq z \leq 1$.
21. By transforming to a triple integral evaluate, $\iiint_S x^3dydz + x^2zdxdy$ where S is the closed surface bounded by the planes $x = 0, x = 6$ and cylinder $y^2 + z^2 = a^2$.
22. Apply divergence theorem to evaluate $\iiint_S (x + z)dydz + (y + z)dzdx + (x + y)dxdy$, where S is the surface of the sphere $x^2 + y^2 + z^2 = 4$.